

Written HW 3 — Full Solutions

Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 23.

$\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$, and $g = 9.8 \text{ m/s}^2$ throughout.

23.3 (2 pts)

Two point charges, $q_A = +3.0 \text{ nC}$ and $q_B = -3.0 \text{ nC}$, sit one above the other with a 5.0 cm gap on each side of the point marked by the dot in Figure EX23.3: q_A is 5.0 cm above the dot's height, q_B is 5.0 cm below it, and the dot itself is 5.0 cm to the side of the line joining the two charges. What are the strength and direction of the electric field at the dot? Specify the direction as an angle cw from horizontal.

MODEL: Treat both charges as point charges. The field at the dot is the vector sum of the field from q_A alone and the field from q_B alone.

VISUALIZE: Put the midpoint between the charges at the origin, q_A at (0, 5.0 cm), q_B at (0, -5.0 cm), and the dot at (5.0 cm, 0). Both charges are the same distance from the dot, along lines at 45° to the vertical. q_A is positive, so its field points away from q_A , down and to the right. q_B is negative, so its field points toward q_B , down and to the left. The horizontal components point opposite ways and will cancel; the vertical components both point down and will add.

SOLVE: Each charge is a distance $r = \sqrt{(5.0 \text{ cm})^2 + (5.0 \text{ cm})^2} = 7.07 \text{ cm} = 0.0707 \text{ m}$ from the dot, so

$$E_A = E_B = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} = \frac{(8.99 \times 10^9)(3.0 \times 10^{-9})}{(0.0707 \text{ m})^2} = 5.39 \times 10^3 \text{ N/C}.$$

Each field makes a 45° angle with the vertical, so each has horizontal component $E \cos 45^\circ$ and vertical component $E \sin 45^\circ$, both equal in magnitude since $\sin 45^\circ = \cos 45^\circ$. The horizontal components are equal and opposite (one points right, one points left), so they cancel exactly. The vertical components both point downward and add:

$$E_{\text{net}} = 2E_A \sin 45^\circ = 2(5.39 \times 10^3 \text{ N/C})(0.707) = \boxed{7.6 \times 10^3 \text{ N/C, straight down (90}^\circ \text{ cw from horizontal)}}.$$

REVIEW: The horizontal cancellation isn't a coincidence: the dot is equidistant from both charges and the charges have equal and opposite magnitude, so whatever field q_A contributes sideways, q_B contributes the exact opposite. Only the vertical components, which both point the same way (from + toward -), survive. That the net field ends up parallel to the line joining the charges is a general feature of any symmetric point on a dipole's perpendicular bisector.

23.4 (2 pts)

$q_A = +3.0$ nC sits at the same height as, and 5.0 cm to the left of, the dot marked in Figure EX23.4. $q_B = -6.0$ nC sits 10 cm directly below q_A . What are the strength and direction of the electric field at the dot? Specify the direction as an angle cw from horizontal.

MODEL: Treat both charges as point charges and add their fields as vectors.

VISUALIZE: Put q_A at the origin, the dot at (5.0 cm, 0), and q_B at (0, -10 cm). q_A is positive and directly to the left of the dot, so its field at the dot points straight in the $+x$ direction. q_B is negative and down and to the left of the dot, so its field at the dot points down and to the left, toward q_B .

SOLVE: q_A is a distance $r_A = 5.0$ cm = 0.050 m from the dot:

$$E_A = \frac{1}{4\pi\epsilon_0} \frac{q_A}{r_A^2} = \frac{(8.99 \times 10^9)(3.0 \times 10^{-9})}{(0.050)^2} = 1.08 \times 10^4 \text{ N/C, in } +\hat{x}.$$

q_B is a distance $r_B = \sqrt{(5.0 \text{ cm})^2 + (10 \text{ cm})^2} = 11.18 \text{ cm} = 0.1118 \text{ m}$ from the dot, and the unit vector from the dot toward q_B is $(-5.0, -10)/11.18 = (-0.447, -0.894)$:

$$E_B = \frac{1}{4\pi\epsilon_0} \frac{|q_B|}{r_B^2} = \frac{(8.99 \times 10^9)(6.0 \times 10^{-9})}{(0.1118)^2} = 4.32 \times 10^3 \text{ N/C,} \quad \vec{E}_B = (-1.93, -3.86) \times 10^3 \text{ N/C.}$$

Adding components, $\vec{E}_A = (1.08 \times 10^4, 0) \text{ N/C}$, so

$$\vec{E}_{\text{net}} = (8.86, -3.86) \times 10^3 \text{ N/C} \Rightarrow E_{\text{net}} = \sqrt{8.86^2 + 3.86^2} \times 10^3 \text{ N/C} = \boxed{9.7 \times 10^3 \text{ N/C}},$$

directed $\tan^{-1}(3.86/8.86) = \boxed{23^\circ \text{ below horizontal (cw)}}$, i.e., down and to the right.

REVIEW: q_A is closer to the dot than q_B (5.0 cm vs. 11.2 cm) and q_A 's field points straight along $+\hat{x}$ with nothing to cancel it, so it's no surprise that q_A dominates the result: the net field is close to E_A 's value and tilted only mildly toward q_B 's direction.

23.8 (3 pts)

A 10-cm-long thin glass rod uniformly charged to +10 nC and a 10-cm-long thin plastic rod uniformly charged to -10 nC are placed side by side, 4.0 cm apart. What are the electric field strengths E_1 to E_3 at distances 1.0 cm, 2.0 cm, and 3.0 cm from the glass rod, along the line connecting the midpoints of the two rods?

MODEL: Each rod is a finite uniformly charged line. On the line through a rod's own midpoint, perpendicular to the rod, the field of a rod with charge Q and length L is

$$E_{\text{rod}}(y) = \frac{1}{4\pi\epsilon_0} \frac{Q}{y\sqrt{y^2 + (L/2)^2}},$$

directed away from the rod if $Q > 0$, toward it if $Q < 0$. All three points lie on this line for *both* rods at once (they're between the rods), so the total field is the vector sum of the two rods' contributions.

VISUALIZE: Put the glass rod's center at $x = 0$ and the plastic rod's center at $x = 4.0$ cm. A point at distance y_1 from the glass rod (measured from $x = 0$) is then a distance $y_2 = 4.0 \text{ cm} - y_1$ from the plastic rod. Glass is positive, so its field pushes away from itself, toward the plastic rod ($+x$). Plastic is negative, so its field pulls toward itself, which from a point between the rods is also toward $+x$. Both fields point the same way at every point between the rods, so the strengths simply add.

SOLVE: With $L = 0.10$ m so $L/2 = 0.050$ m, and $Q = 1.0 \times 10^{-8}$ C for each rod (same magnitude, opposite sign):

$$E_1 : y_1 = 1.0 \text{ cm}, y_2 = 3.0 \text{ cm} \Rightarrow E_{\text{glass}} = 1.76 \times 10^5, E_{\text{plastic}} = 5.14 \times 10^4 \Rightarrow E_1 = \boxed{2.3 \times 10^5 \text{ N/C}}$$

$$E_2 : y_1 = y_2 = 2.0 \text{ cm} \Rightarrow E_{\text{glass}} = E_{\text{plastic}} = 8.35 \times 10^4 \Rightarrow E_2 = \boxed{1.7 \times 10^5 \text{ N/C}}$$

$$E_3 : y_1 = 3.0 \text{ cm}, y_2 = 1.0 \text{ cm} \Rightarrow E_{\text{glass}} = 5.14 \times 10^4, E_{\text{plastic}} = 1.76 \times 10^5 \Rightarrow E_3 = \boxed{2.3 \times 10^5 \text{ N/C}}$$

all three directed from the glass rod toward the plastic rod.

REVIEW: E_1 and E_3 come out identical. That has to be true: the point 1.0 cm from glass (3.0 cm from plastic) is the mirror image of the point 3.0 cm from glass (1.0 cm from plastic), with the roles of "near rod" and "far rod" simply swapped between two rods that carry equal and opposite charge of equal magnitude. The field is *smaller* exactly in the middle than near either rod: moving toward a rod, that rod's own contribution grows faster than the other rod's contribution shrinks.

23.10 (2 pts)

A small glass bead, charged to $+8.0 \text{ nC}$, is in the plane that bisects a thin, uniformly charged, 10-cm-long glass rod and is 4.0 cm from the rod's center. The bead is repelled from the rod with a force of $720 \text{ }\mu\text{N}$. What is the total charge on the rod?

MODEL: The bead sits on the rod's perpendicular bisector, so its field there is $E = \frac{1}{4\pi\epsilon_0} Q_{\text{rod}} / (y\sqrt{y^2 + (L/2)^2})$. The force on the bead is $F = qE$.

VISUALIZE: The bead is *repelled*, so the rod's charge must have the same sign as the bead's: the rod is positively charged.

SOLVE: First find the field the bead sits in from the force it feels:

$$E = \frac{F}{q} = \frac{7.20 \times 10^{-4} \text{ N}}{8.0 \times 10^{-9} \text{ C}} = 9.0 \times 10^4 \text{ N/C}.$$

With $y = 0.040 \text{ m}$ and $L/2 = 0.050 \text{ m}$, $y\sqrt{y^2 + (L/2)^2} = (0.040)\sqrt{0.0016 + 0.0025} = 2.56 \times 10^{-3} \text{ m}^2$, so

$$Q_{\text{rod}} = \frac{Ey\sqrt{y^2 + (L/2)^2}}{1/(4\pi\epsilon_0)} = \frac{(9.0 \times 10^4)(2.56 \times 10^{-3})}{8.99 \times 10^9} = \boxed{+26 \text{ nC}}.$$

REVIEW: 26 nC on a 10-cm rod is a linear density of about 2.6 nC/cm, roughly comparable to the point charges used elsewhere in this chapter's problems, so the answer is a believable, everyday-lab-bench amount of charge rather than something wildly large or small.

23.12 (2 pts)

The electric field 2.0 cm from a very long charged wire is 1000 N/C, directed toward the wire. What is the charge (in nC) on a 5.0-cm-long segment of the wire?

MODEL: Far from its ends, a long charged wire produces the field of an infinite line of charge, $E = \lambda/(2\pi\epsilon_0 r)$, independent of the wire's total length.

VISUALIZE: The field points *toward* the wire, which is what a negative line charge does (it attracts a positive test charge in toward itself). So the wire is negatively charged.

SOLVE: Solve for the linear charge density:

$$\lambda = 2\pi\epsilon_0 Er = 2\pi(8.85 \times 10^{-12})(1000 \text{ N/C})(0.020 \text{ m}) = 1.11 \times 10^{-9} \text{ C/m}.$$

The charge on a 5.0-cm = 0.050 m segment is then

$$Q = \lambda L = (1.11 \times 10^{-9} \text{ C/m})(0.050 \text{ m}) = 5.6 \times 10^{-11} \text{ C} = \boxed{-0.056 \text{ nC}}$$

(negative, since the wire itself is negatively charged).

REVIEW: The field formula for a line of charge doesn't care how long the wire is, only how densely charged it is and how far away you're standing; the 5.0-cm length only enters at the very last step, to convert the density into an actual amount of charge on that particular stretch of wire.

23.19 (3 pts)

Two very large sheets of plastic face each other, separated by distance d , as shown in Figure EX23.19. One sheet (on the left) has been rubbed to a uniform surface charge density $\eta_1 = -\eta_0$; the other (on the right) has surface charge density $\eta_2 = +3\eta_0$. What are the electric field vectors at points 1 (outside, to the left of both sheets), 2 (midway between the sheets), and 3 (outside, to the right of both sheets)?

MODEL: Each sheet, treated as an infinite plane of charge, produces a field of constant magnitude $\eta/(2\epsilon_0)$ on each side, independent of distance from that sheet: pointing away from the sheet if $\eta > 0$, toward it if $\eta < 0$. The total field at any point is the vector sum of both sheets' contributions.

VISUALIZE: Put the negative sheet at $x = 0$ and the positive sheet at $x = d$, with $+x$ pointing to the right. Sheet 1 ($-\eta_0$) points its field toward itself everywhere: in $+\hat{x}$ to its left (point 1) and in $-\hat{x}$ everywhere to its right (points 2 and 3). Sheet 2 ($+3\eta_0$) points its field away from itself everywhere: in $-\hat{x}$ everywhere to its left (points 1 and 2) and in $+\hat{x}$ to its right (point 3).

SOLVE: Each sheet contributes magnitude $\eta_0/(2\epsilon_0)$ (sheet 1) or $3\eta_0/(2\epsilon_0)$ (sheet 2). Adding the two contributions, with sign for direction ($+\hat{x}$ positive):

$$\text{Point 1 (left of both): } \frac{\eta_0}{2\epsilon_0} - \frac{3\eta_0}{2\epsilon_0} = -\frac{\eta_0}{\epsilon_0} \Rightarrow \vec{E}_1 = \boxed{\frac{\eta_0}{\epsilon_0}, \text{ pointing left, away from the sheets}}$$

$$\text{Point 2 (between): } -\frac{\eta_0}{2\epsilon_0} - \frac{3\eta_0}{2\epsilon_0} = -\frac{2\eta_0}{\epsilon_0} \Rightarrow \vec{E}_2 = \boxed{\frac{2\eta_0}{\epsilon_0}, \text{ pointing left, from sheet 2 toward sheet 1}}$$

$$\text{Point 3 (right of both): } -\frac{\eta_0}{2\epsilon_0} + \frac{3\eta_0}{2\epsilon_0} = +\frac{\eta_0}{\epsilon_0} \Rightarrow \vec{E}_3 = \boxed{\frac{\eta_0}{\epsilon_0}, \text{ pointing right, away from the sheets}}$$

Note d never enters: an infinite sheet's field doesn't depend on distance from it.

REVIEW: The two outside points have equal field strength, and the field between the sheets is exactly double that. This is the same pattern as a parallel-plate capacitor (a much stronger field trapped between the plates than leaks out around them) even though here the two charge densities aren't equal and opposite (η_0 vs. $3\eta_0$): the ratio between the "inside" and "outside" field strength is set purely by how the two contributions add or cancel, not by the sheets matching each other.

23.25 (4 pts)

Figure EX23.25 shows a 1.5 g ball hanging from a string inside a parallel-plate capacitor made from $12\text{ cm} \times 12\text{ cm}$ electrodes, charged to $\pm 75\text{ nC}$. The string hangs at 15° from vertical, deflected toward the negative plate. What is the charge on the ball, in nC?

MODEL: Treat the capacitor's field as that of an ideal parallel-plate capacitor, uniform between the plates: $E = Q_{\text{plate}}/(\epsilon_0 A)$. The ball hangs in equilibrium under three forces: gravity, string tension, and the electric force qE .

VISUALIZE: The string is deflected toward the negative plate. The field inside the capacitor points from the positive plate toward the negative plate. A ball that gets pushed *along* the field, toward the negative plate, must be positively charged.

SOLVE: The plate area is $A = (0.12\text{ m})^2 = 0.0144\text{ m}^2$, so the field strength is

$$E = \frac{Q_{\text{plate}}}{\epsilon_0 A} = \frac{75 \times 10^{-9}\text{ C}}{(8.85 \times 10^{-12})(0.0144\text{ m}^2)} = 5.89 \times 10^5\text{ N/C}.$$

For the hanging ball, the string's angle from vertical is set by the ratio of the horizontal electric force to the vertical weight, $\tan \theta = F_{\text{elec}}/(mg)$, so

$$F_{\text{elec}} = mg \tan \theta = (1.5 \times 10^{-3}\text{ kg})(9.8\text{ m/s}^2) \tan 15^\circ = 3.94 \times 10^{-3}\text{ N}.$$

Then

$$q = \frac{F_{\text{elec}}}{E} = \frac{3.94 \times 10^{-3}\text{ N}}{5.89 \times 10^5\text{ N/C}} = \boxed{+6.7\text{ nC}}.$$

REVIEW: A 15° deflection is fairly gentle, so it makes sense that the electric force (3.9 mN) comes out smaller than the ball's weight ($mg \approx 14.7\text{ mN}$) rather than dwarfing it: $\tan 15^\circ \approx 0.27$ is a modest ratio, not a dramatic one. If the deflection had instead been described toward the positive plate, the same method would give the same magnitude of charge, just negative.